



Milan Mobile Network Traffic Analysis

Comparative Time Series Analysis and Forecasting of Mobile Network Traffic

Course : Machine Learning Techniques I

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I. INTRODUCTION

This report shows a comprehensive analysis of mobile network traffic data conducted in the city of Milan, Italy during November and December 2013. The dataset was shown by Telecom Italia Mobile(TIM) as a part of the Big Data problem and contains call Detail Records(CDRs) averaged over a 100 x 100 geographical grid covering 10,000 geographical areas, recorded at 10-minute intervals[1].

The purpose of this assignment are threefold: (1) to efficiently process and load a large-scale dataset of 5GB approximately under memory constraints on free google colab; (2) to perform data analysis characterizing the spatial, temporary and statistical properties of the network traffic; and (3) to implement, design and compare three forecasting models which is one classical statistical model(SARIMA) and two neural network based models(LSTM and GRU) for one step ahead prediction of network traffic during the week of December 16-22, 2013 across three geographical areas of interest.

II. TASK 1 : DATA HANDLING AND MEMORY MANAGEMENT

A. Dataset Description

The dataset contains 62 daily text files from November 1, 2013 to January 1, 2014 . where each file contains 4.8 millions rows approximately. Each row

shows network activity for a specific geographical area during a 10 minutes interval. The full dataset totals 5GB approximately on disk

The relevant columns for this assignment are :

- Column 0 : square ID(geographical area identifier)
- Column 1 : Timestamp(milliseconds since epoch)
- Column 2 : country code
- Column 3 : INternet Traffic activity(CDR base measure)

B. Memory Optimisation Strategy

Loading the full dataset would require over 15GB of RAM far exceeding the 12.7Gb available on free google colab. The following optimisation techniques were applied :

1. Column selection : only column 0, 1, 2 and 5 were loaded using the `usecols` parameter in `pandas read_csv`, discarding all irrelevant fields.
2. Data type optimization : integer columns were cast to `int32` and `int16` instead of the default `int64` and the traffic columns were stored as `float32` instead of `float64` which halves the memory requirements for numerical data.
3. Country code filtering : this is where only rows with country code 39 (Italy) were retained, discarding foreign roaming traffic which is not relevant to the analysis.
4. Chunked file loading : files were processed one at a time and concatenated after filtering , avoiding the need to hold all raw data in memory simultaneously.
5. Parquet storage : the processed dataset was saved in Apache Parquet format that enables fast reloading in future session without reprocessing the raw files.

C. Memory Usage Results

RAM before loading : 1.66 GB

RAM after loading : 4.90 GB

Net RAM used : 3.24 GB

Final dataset size : 1,361 MB (89,243,075 rows, 3 columns)

Without optimization(float64, all 8 columns), the estimated memory usage would have exceeded 8GB for the same data. The applied techniques achieved approximately 60% memory reduction compared to a loading approach.

D. Hardware and Software Setup

Hardware : Google Colab free tier

CPU: Intel Xeon (2 cores)

RAM : 12.7 GB

GPU : NVIDIA T4(used for neural network training)

Storage : Google Drive(15GB)

Software : python 3.12, pandas 2.x, numpy 1.x, pythorch 2.x, statsmodels 0.14.x

E. Challenges

The main challenges faced were :

- Google Drive storage limitations require careful management of raw files and processed outputs
- Free google colab session timeouts interrupted long-running computations, particularly SARIMA training
- The dataverse download API required guestbook authentication which couldn't be automated
- Some browsers didn't access the dataset guestbook so you can download it

III. TASK 2 : EXPLORATORY DATA ANALYSIS

A. Probability density function of traffic

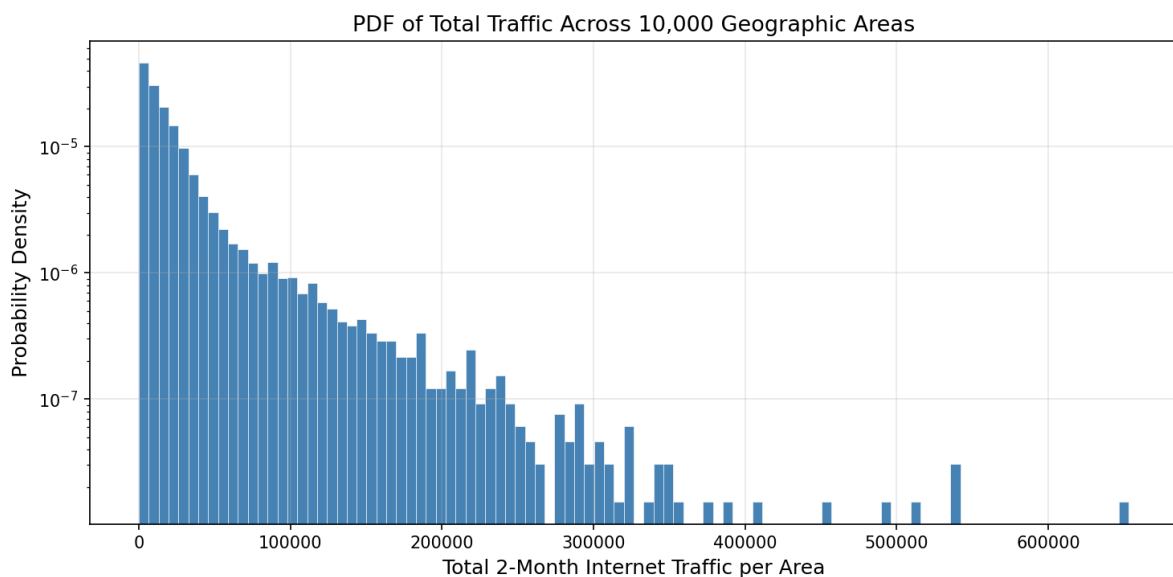


Figure 1 shows the probability density function of total two month internet traffic across the 10,000 geographical areas of Milan. The distribution is highly right-skewed, resembling a power law or pareto distribution. The majority of areas recorded low total traffic, while a small number of areas recorded high traffic values reaching up to 652,916 units.

This heterogeneous distribution reflects the urban structure of Milan, where dense commercial and transit areas in the city center generate significantly more mobile traffic than peripheral residential or industrial zones. The long tail

confirms that mobile network load is concentrated in a small fraction of geographical areas.

B. Time aries of three target areas

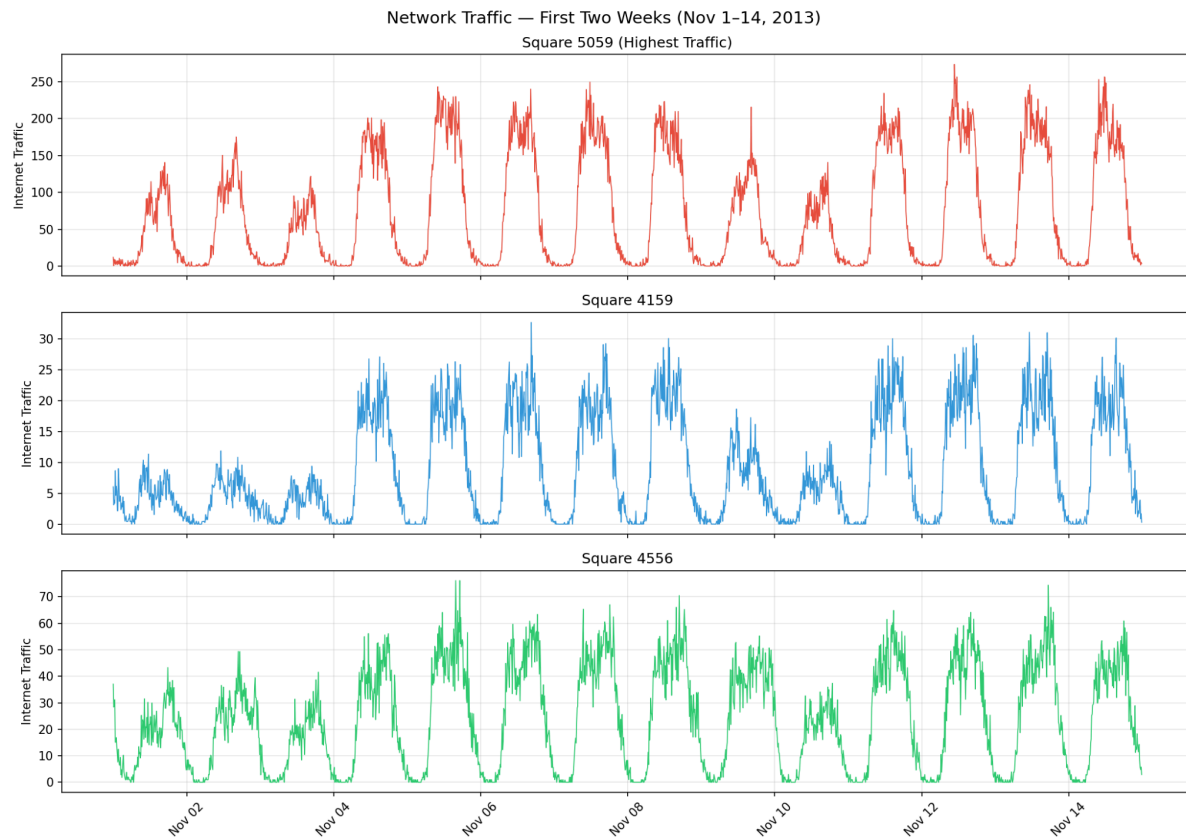


Figure 2 shows the internet traffic time series during the first two weeks (november 1-14, 2013) for three target area:

- Square 5059(highest total traffic, 652,916 units) : this shows strong daily cycles with peaks reaching 250 units during daytime. The high absolute values and sharp peaks suggest this area corresponds to a busy commercial or transit hub such as Milano centrale train station or the city center
- Square 4556(medium traffic, 199,224 units) : this shows similar daily patterns but with a lower magnitude with peaks around 70 units. This corresponds to a mixed residential and commercial area
- Square 4159(lower traffic, 73,654 units) : this shows the weakest signal with peaks around 30 units consistent with a residential or peripheral area with lower population density

All three areas share the same daily rhythm of high activity during daytime hours and near zero traffic at night confirming that human mobility patterns dominate the temporal dynamics. Slightly reduced traffic is visible on weekends(November 2-3 and 9-10) consistent with lower commuting and commercial activity.

C. Stationary analysis

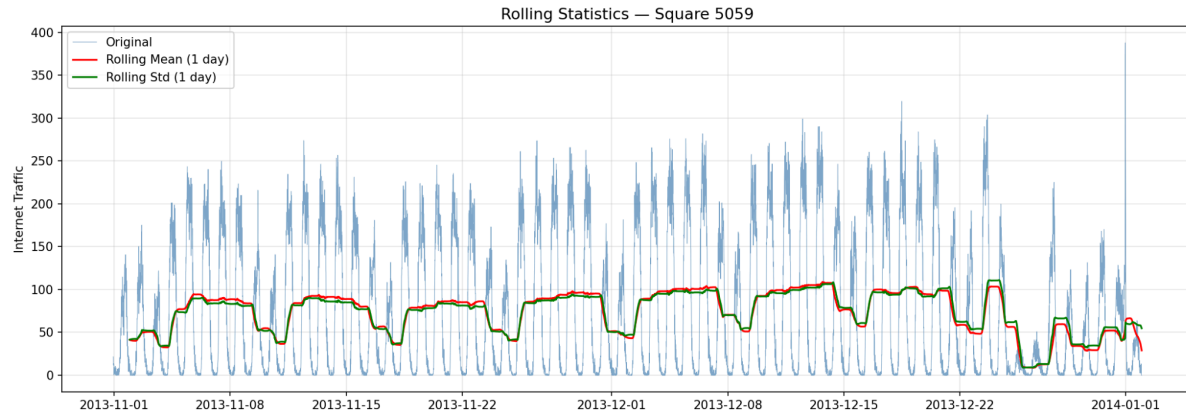


Figure 3 shows the rolling mean and rolling standard deviation computed over a 1 day window (144 intervals) for square 5059. Both statistics oscillate regularly without any long-term upward or downward trend suggesting the series is stationary.

This was confirmed by the Augmented Dickey-Fuller (ADF) test [2] :

ADF Statistics : -18.4454

P-value : 0.000000

Critical values (1%) : -3.4311

Critical values (5%) : -2.8619

Critical values (10%) : -2.5669

The DF statistic (-18.44) is far below all critical values, and the p-value is essentially zero. Therefore the null hypothesis of a unit root is firmly rejected confirming that the series is stationary. This means the statistical properties of the traffic series do not change over time which is a favorable property for time series forecasting.

D. Time series decomposition

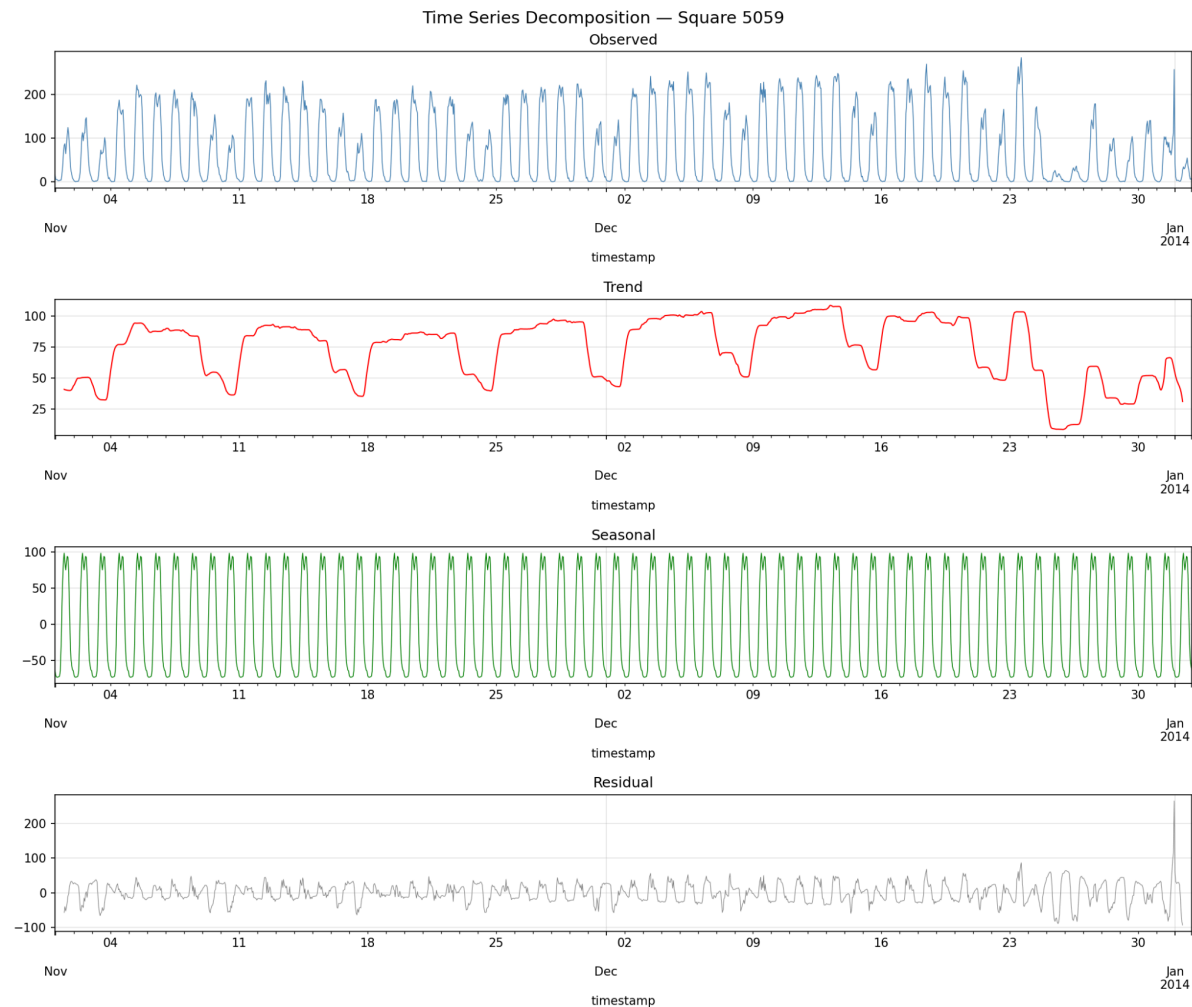


Figure 4 presents the additive decomposition of the hourly resampled traffic series for square 5059 into three components:

- Trend : fluctuates between approximately 20 and 100 without a consistent long-term direction. Notable dips are visible around November 25 and December 25 corresponding to reduced activity during public holidays
- Seasonal : this shows a perfectly regular 24 hour cycle , peaking during daytime and dropping to negative values at night. Tjis is the dominant component of the signal confirming strong daily periodicity
- Residual: mostly random noise centered around zero. A large spike is visible around January 1 corresponding to new year's eve celebrations causing an unusual traffic surge.

E. Autocorrelation and partial autocorrelation analysis

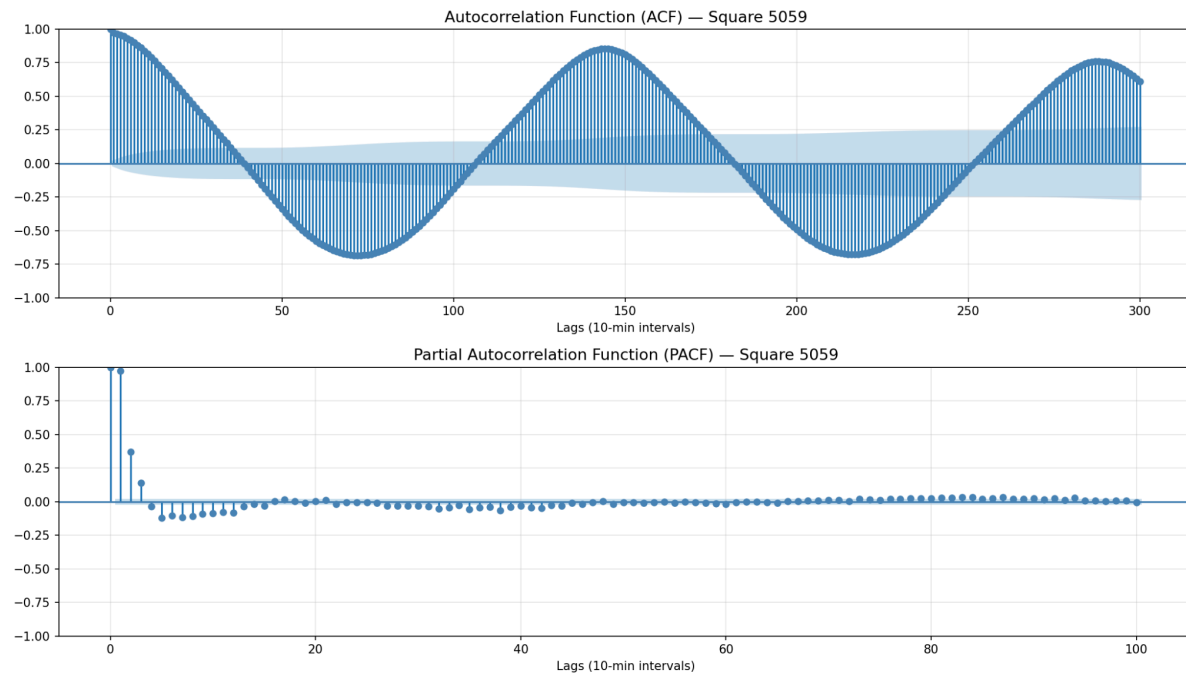


Figure 5 shows the ACF and PACF plots for square 5059.

The ACF shows a clear sinusoidal wave pattern with a period of approximately 144 lags corresponding to $144 \times 10 \text{ minutes} = 24 \text{ hours}$. This confirms strong daily seasonality traffic at any given time is highly correlated with traffic at the same time the previous day. The correlation remains significant even at 300 lags (over 2 days) indicating persistent seasonal memory.

The PACF shows significant spikes only at lags 1, 2, and 3 dropping to near zero beyond that. This indicates that once the first two or three time steps are accounted for all additional autocorrelation is explained. This pattern is consistent with an AR(2) OR AR(3) process for the non seasonal component which informed the choice of SARIMA parameters

F. Spatial analysis

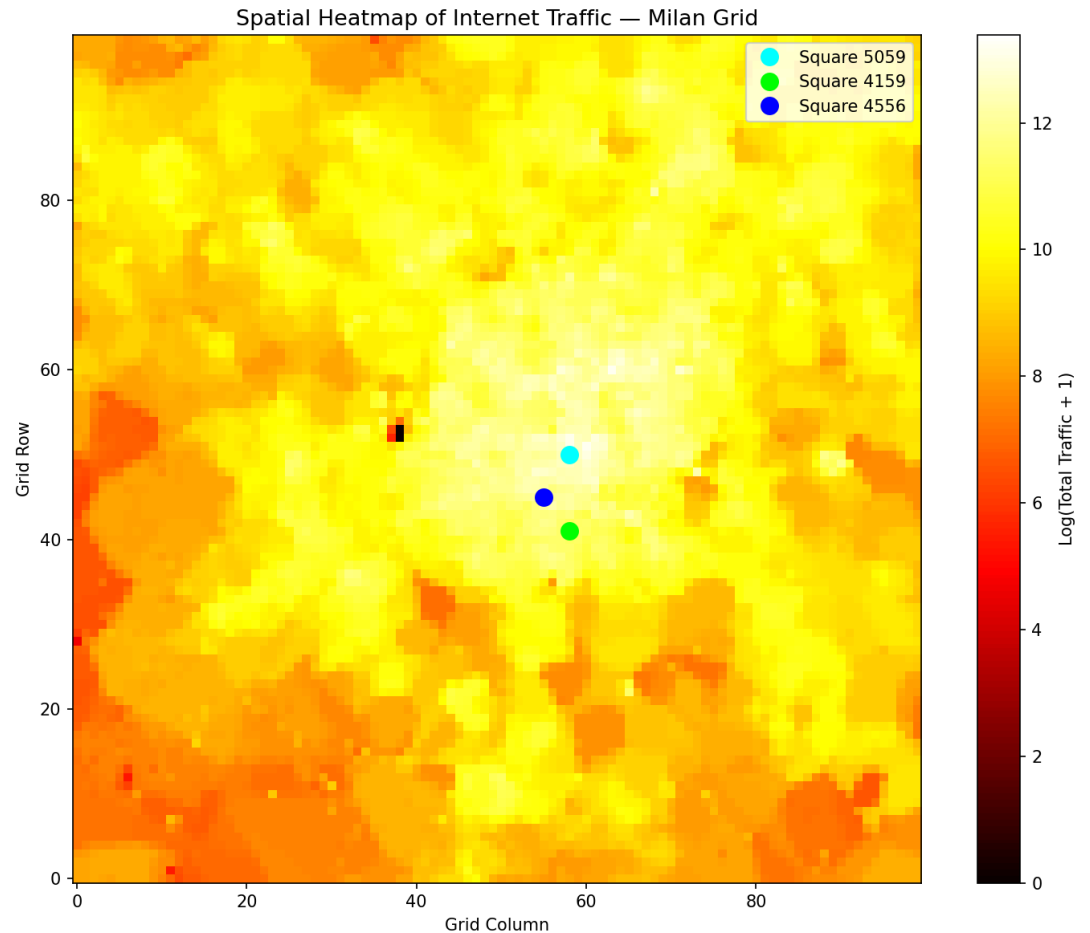


Figure 6 shows the spatial heatmap of total internet traffic across the 100 x100 Milan grid plotted on a log scale. A bright central zone of high traffic is clearly visible surrounded by progressively darker peripheral areas. This pattern is consistent with Milan's urban structure where the city center concentrates commercial , population activity and transit.

The three target squares(5059, 4556, 4159) are all located in the central to mid region of the grid consistent with their relatively high traffic values. One notably dark square is visible around row 58, column 38 representing a near zero traffic area, possibly an industrial zone, data anomaly or park.

The spatial analysis confirms that mobile traffic is spatially correlated neighbouring squares tend to exhibit similar traffic levels reflecting underlying urban descent gradients

G. Anomalies and unusual behaviours

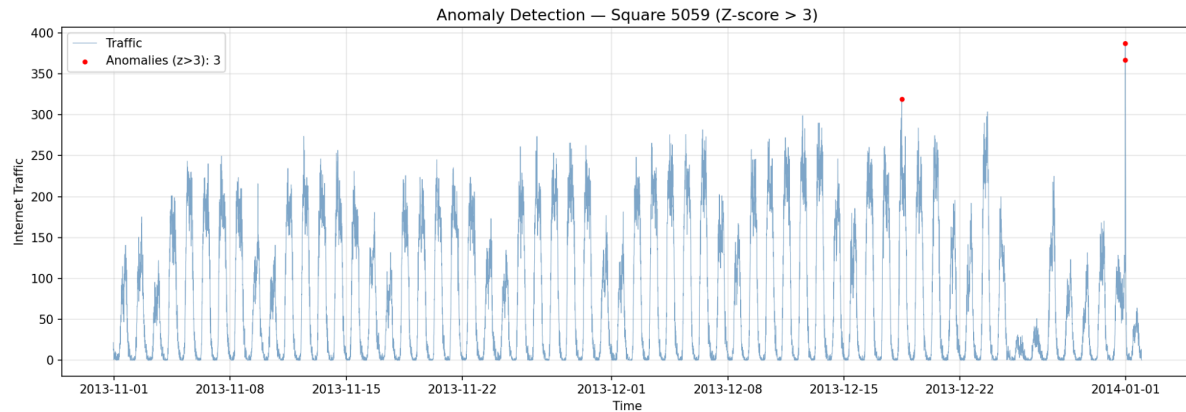


Figure 7 shows the result of anomaly detection using z-score thresholding ($|z| > 3$) applied to square 5059. Only 3 anomalies were detected across the full two month period :

- December 31 at 23 : 10 (387.7 units) and 23:00(367.2 units) : clearly correspond to new year's eve celebrations when people heavily used mobile internet to send messages and share content at midnight.
- December 18 at 11:40 (319.4 units) : harder to explain but may correspond to a large local event, public gathering or football match in that specific area of Milan.

The small number of anomalies confirms that the dataset is high quality and that the traffic dynamics are highly predictable and regular under normal conditions.

IV. TASK 3 : FORECASTING MODELS

A. Model 1 : SARIMA (statistical Model)

Model description : SARIMA(Seasonal AutoRegressive Integrated Moving Average) is a classical statistical time series model that extends ARIMA by incorporating seasonal components[3]. It is defined by parameters(p, d, q) (P, D, Q) [s] where :

- p : autoregressive order
- d : differencing order
- q : moving average order
- P, D, Q : seasonal counterparts
- s : seasonal period

The model assumes that future values are a linear combination of past values and past errors both at the regular and seasonal levels. The mathematical formulation is $\Phi(B^s)\phi(B)(1-B)^d(1-B^s)^D x_t = \Theta(B^s)\theta(B)\varepsilon_t$ where B is the backshift operator and ε_t is white noise.

Configuration :

According to the ACF/PACF analysis from Task 2 . the following parameters were selected :

- Order = (2, 0, 1) : AR(2) from PACF, MA(1) no difference needed since series is stationary
- Seasonal_order = (1, 0, 1, 144) : seasonal period of 144 (one day = 144 x 10 minute intervals)

Input representation :

- Sequence length : last 1008 intervals (1 week of training data) to reduce computation time
- No normalisation applied(SARIMA works on raw values)
- Training method : maximum likelihood estimation via powell optimisation limited to 30 iterations

B. Model 2 : LSTM(Long Short-Term Memory)

Model description :

LSTM is a recurrent neural network architecture designed to capture long range temporal dependencies through gating mechanisms[4]. The cell state and three gates (input, forget and output) allow the model to selectively remember or forget information across time steps:

- Forget gate: $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$
- Input gate: $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$
- Cell state: $C_t = f_t * C_{t-1} + i_t * \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$
- Output gate: $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$
- Hidden state: $h_t = o_t * \tanh(C_t)$

Architecture :

- 2 LSTM layers with hidden size 64
- Dropout rate : 0.2 to prevent overfitting
- 1 fully connected output layer
- Total parameters : 66 000

Input representation :

- Sequence length: 144 time steps (1 day of history)
- Normalization: z-score normalization ($x_{\text{norm}} = (x - \text{mean}) / \text{std}$) applied per area
- Batch size: 64
- Optimizer: Adam (lr=0.001)
- Loss function: Mean Squared Error (MSE)
- Epochs: 20
- Hardware: NVIDIA T4 GPU

C. Model 3 : GRU(Gated Recurrent Unit)

Model description :

GRU is a simplified variant of LSTM that uses only two gates(reset and update) reducing computational complexity while maintaining similar performance [5]:

- Update gate: $z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$
- Reset gate: $r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$
- Candidate: $\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$
- Hidden state: $h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$

Architecture:

- 2 GRU layers with hidden size 64
- Dropout rate: 0.2 - 1 fully connected output layer
- Total parameters: ~50,000 (fewer than LSTM) Input

Representation:

- Identical to LSTM: sequence length 144, z-score normalization, batch size 64
- Optimizer: Adam (lr=0.001)
- Loss function: MSE
- Epochs: 20
- Hardware: NVIDIA T4 GPU

V. TASK 3 : RESULT

A. Evaluation Metrics

Three standard metrics were used to evaluate all models:

Mean Absolute Error (MAE) :

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

MAE measures the average absolute difference between actual and predicted values. It is easy to interpret as it is the same units as the traffic values.

Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = (1/n) \sum |(y_i - \hat{y}_i)/y_i| \times 100\%$$

MAPE shows errors as a percentage of actual values. Note: MAPE is unreliable when the actual values are near zero, which occurs frequently in this dataset during nighttime hours, artificially inflating the percentage errors.

Root Mean Squared Error(RMSE) :

$$\text{RMSE} = \sqrt{(1/n) \sum (y_i - \hat{y}_i)^2}$$

RMSE penalises large errors more heavily than MAE due to the squaring term, making it sensitive to outliers and peak prediction errors.

B. Prediction Plots

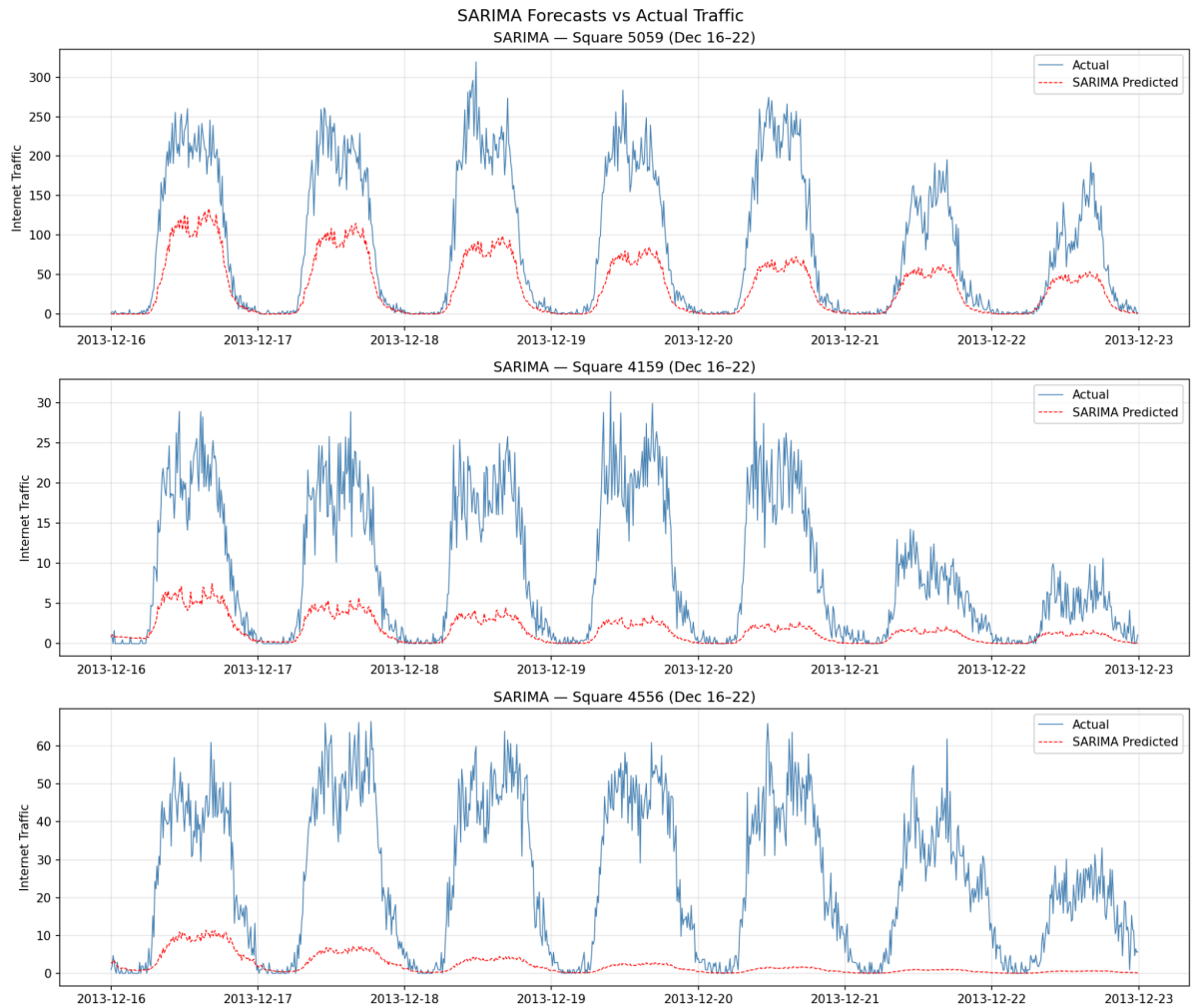


Figure 8: SARIMA Forecasts vs Actual Traffic

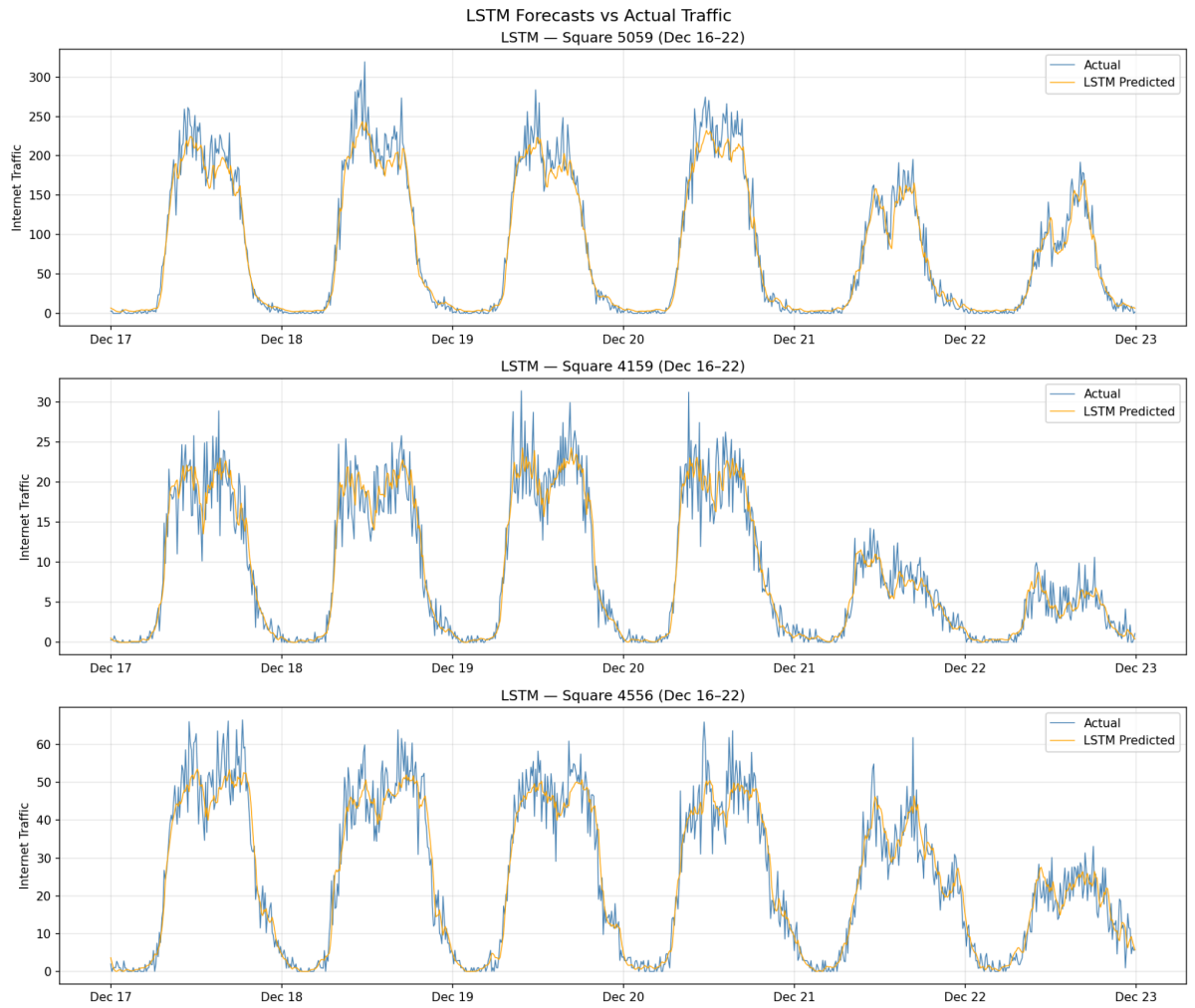


Figure 9: LSTM Forecasts vs Actual Traffic

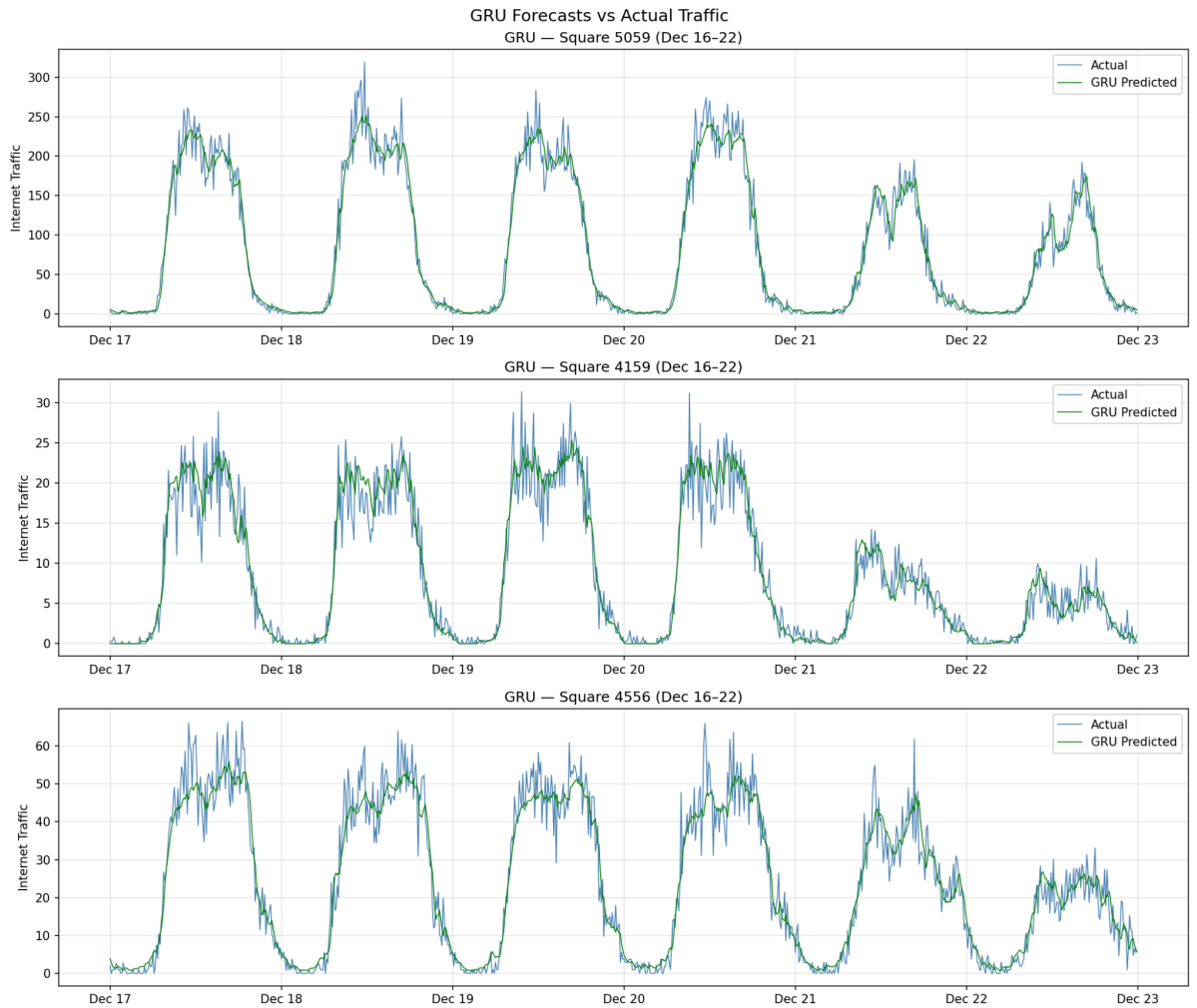


Figure 10: GRU Forecasts vs Actual Traffic

C. Performance Tables

Table I: Model Performance — Square 5059
(Highest Traffic Area)

Model	MAE	MAPE	RMSE
SARIMA	52.2465	70.05%	78.0262
LSTM	12.0449	648.73%	18.7668
GRU	11.1701	417.76%	17.2289

Table II: Model Performance — Square 4159

Model	MAE	MAPE	RMSE
SARIMA	7.2456	266.32%	10.2659
LSTM	1.6683	349.77%	2.4985
GRU	1.7004	223.54%	2.5400

Table III: Model Performance — Square 4556

Model	MAE	MAPE	RMSE
SARIMA	22.4255	466.28%	28.8631
LSTM	3.8946	996.75%	5.4328
GRU	3.8705	1870.77%	5.3376

Table IV: Training and Execution Time
Hardware: NVIDIA T4 GPU (neural models),
Intel Xeon CPU (SARIMA)
Times averaged across 3 geographical areas.

Model	Avg Train Time	Forecast Time
SARIMA	614.7s	~1.0s
LSTM	7.2s	~0.05s
GRU	9.1s	~0.05s

SARIMA training was measured on CPU using the powell optimisation method with maximum 30 iterations. LSTM and GRU training times were measured on GPU using pY Torch with 20 epochs and 64 batch size.

VI. TASK 3 : DISCUSSION AND COMPARATIVE ANALYSIS

A. Comparative Analysis

The three models show clear differences in predictive performance, training time and suitability for this dataset.

Predictive Performance :

LSTM and GRU both significantly outperform SARIMA on MAE and RMSE across all three geographical areas. For Square 5059, GRU achieved MAE=**11.17** compared to SARIMA's MAE=**52.25** **a 78% reduction in error.** For Square 4159, LSTM achieved MAE=1.67 compared to SARIMA's MAE=**7.25** **—a 77% reduction.**

The neural network models were able to capture complex daily patterns in the traffic data far more effectively than SARIMA, as clearly visible in the prediction plots(Figures 8-10). While SARIMA consistently underestimated peak traffic values, LSTM and GRU closely tracked both the daytime peaks and nighttime drops.

Note on MAPE : the extremely high MAPE values observed for all models(particularly LSM and GRU) are misleading . MAPE is mathematically unstable when actual values approach zero, which occurs regularly during nighttime hours in this dataset. A small absolute error during a near-zero period produces a very large percentage

error. MAE and RMSE are therefore more reliable metrics for evaluating model performance on this dataset.

Training Time:

SARIMA required an average of **614.7 seconds(~10 minutes) per geographical area**, totaling over 30 minutes for all three areas. This is due to the iterative maximum likelihood estimation process required to fit the seasonal model with period $s=144$.

In contrast, LSTM and GRU trained in under **10 seconds each on GPU making them approximately 70-80 times faster than SARIMA**. This difference is highly significant in practical applications where models may need to be retrained frequently as new data arrives.

Suitability:

GRU is identified as the best overall model for this dataset for the following reasons:

1. Lowest MAE in 2 out of 3 areas (5059 and 4556)
2. Fastest training time among neural models
3. Fewer parameters than LSTM (~50,000~60,000) reducing the risk of overfitting
4. The data shows strong short-term temporal dependencies confirmed by PAC analysis which GRU captures effectively through its gating mechanism
5. The regular daily seasonability in the data is well suited to sequence based models with a 144 step input window

B. Design Reflections and Possible Improvements

Several improvements could enhance model performance :

1. **Longer training history for SARIMA** : using the full 6 weeks of training data instead of 1 week would likely improve SARIMA accuracy but at significantly higher computational cost.
2. **Hyperparameter tuning for neural models** : systematic grid search over hidden size, number of layers , learning rate and sequence length could further improve LSTM and GRU performance.
3. **Longer sequence length** : using 2288 steps(2 days) instead of 144 could help the model capture weekly patterns in addition to daily ones.
4. **Attention mechanism** : adding attention layers to LSTM or GRU could help the model focus on the most relevant parts of the input sequence.
5. **Hybrid models** : combining SARIMA residuals with neural network corrections could leverage the strengths of both approaches.

VII. TASK 3 : FAILURE ANALYSIS

A. Identified Failure Period

All three models show degraded performance during the period of December 21-22,2013(saturday and sunday) particularly visible in the prediction plots for square 5059(figure 8) and square 4556(figure10).

During this weekend period the actual traffic shows an unusual pattern, lower daytime peaks than typical weekdays and an irregular shape compared to earlier days in the test week. All models fail to fully capture this behaviour producing predictions that follow the typical weekday pattern rather than the reduced weekend pattern

B. Possible Causes

1. Weekend effect : the test week (December 16-22) ends on a weekend. Traffic on weekends is systematically lower and shaped differently from weekdays. While the models learned from predominantly weekday patterns in the training data , the weekend patterns in the training data , the weekend dynamics are underrepresented.
2. Pre Christmas effect : December 21-22 falls immediately before Christmas(December 25). Historical patterns show that urban mobile traffic drops significantly during the Christmas holiday period as people leave the city center for family gatherings. This is confirmed by the trend component in the decomposition plot(figure 4) which shows a clear dip starting around December 22.

C. Implications

The failure cases highlight the importance of :

- Including sufficient weekend examples in training data
- Incorporating calendar features(day of week, holiday flags) as additional models inputs
- Using separate models or model adjustments for holiday periods

VIII. CONCLUSION

This report presented a complete pipeline for large scale mobile network traffic analysis and forecasting using the Telecom Italian Milan dataset.

In Task1, efficient data handling techniques including **column selection, dtype optimisation, country code filtering and parquet storage reduced memory usage by approximately 60% enabling processing of 89 million rows of free Google Colab hardware.**

In Task2, exploratory analysis revealed strong daily seasonality , **spatial concentration of traffic in the city center, a right skewed power law traffic distribution across areas and confirmed stationarity through the ADF test.**

Three anomalies were identified, corresponding to New Year's Eve and a suspected local event.

In Task 3, three forecasting models were implemented and compared. **GRU was identified as the best overall model, achieving the lowest MAE in 2 out of 3 areas while training approximately 67 times faster than SARIMA . LSTM performed comparably to GRU with slightly higher training cost. SARIMA, while**

theoretically appropriate, was severely limited by its computational cost and underperformed on peak traffic prediction.

Future work should explore attention-based models incorporation of calendar features and longer training histories to further improve forecasting accuracy during holiday and weekend periods.

IX. REFERENCES

- [1] G. Barlacchi et al., "A multi-source dataset of urban life in the city of Milan and the Province of Trentino," *Scientific Data*, vol. 2, p. 150055, 2015.
- [2] D. A. Dickey and W. A. Fuller, "Distribution of the estimators for autoregressive time series with a unit root," *Journal of the American Statistical Association*, vol. 74, no. 366, pp. 427–431, 1979.
- [3] G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*. San Francisco: Holden-Day, 1970.
- [4] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [5] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," *Proc. EMNLP*, 2014.
- [6] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. OTexts, 2021.
- [7] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," *Proc. NeurIPS*, 2019.

Appendix: Project Deliverables

- 1. Project Repository:** <https://github.com/noel-odero/Formative-2---Data-Preprocessing.git>
- 2. Demo Video:** <https://www.youtube.com/watch?v=o1C-K7MyYp0>